Website Speed Optimization System

Predicting Core Web Vitals and Recommending Website Performance Improvements

Executive Summary

- **Problem**: PageSpeed Insights identifies website performance issues, buy it doesn't estimate the combined LCP impact of improving selected issues.
- **Data**: Collected approximately 19,000 PageSpeed Insights observations
- **Method**: Trained XGBoost regression model to estimate Largest Contentful Paint from performance diagnostic metrics
- **Final Product**: Built an interactive dashboard where users test a URL, adjusted selected metrics by percentile, and recieve an estimate of the potential LCP change.
- **Scope**: The final implementation focuses on LCP. Other Core Web Vitals are future work.

Problem Statement

- Statistics
  - **53%** of mobile users abandon a site that takes more than 3 seconds to load
  - **47%** of users expect any website to load in under 2 seconds
  - **20%** drop in conversions for every 1-second delay in mobile load time
- Slow websites can increase abandonment, reduce conversion, and harm user experience
- Large well-known brands may retain users despite slower performance, but smaller websites are more vulnerable to user abandonment when load times are slow.

Related Work

- Measurement tools
  - Google PageSpeed Insight, Lighthouse, Gtmetrix
- These tools show *what* issues exist, but they do not directly estimate how much improving several issues may change LCP.
- Existing tools measure website performance, but this project will focus on building a dataset and using predictive modeling to estimate performance outcomes and prioritize improvement recommendations.

Project Evolution and Final Scope

- Original ccope: predict multiple Core Web Vitals and generate broad website optimization reocmmendations.
- Challenge discovered: PageSpeed Insight and Lighthouse already provide direct recommendations.
- Refined scope: predict Largest Contentful Paint (LCP) and estimate the potential impact of changes to selected diagnostic metrics.
- Final product: an interactive dashboard for mobile and desktop website testing.
- Future work: extend the method to other Core Web Vitals.

Data Collection

- Started with a large list of websites and expanded each domain to collect relevant subdomains.
- Queried the Google PageSpeed Insights API for each URL using both mobile and desktop strategies.
- Store and process the returned PageSpeed Insight data through a pipeline
- Convert the API responses into one structured observation per tested URL
- Create a modeling dataset containing metrics


```
"https://www.pinks.com/": [
    "https://www.pinks.com/",
    "https://www.pinks.com/invoices",
    "https://www.pinks.com/join-us"
],
"https://www.portosbakery.com/": [
    "https://www.portosbakery.com/",
    "https://www.portosbakery.com/our-story/",
    "https://www.portosbakery.com/community/",
    "https://www.portosbakery.com/gift-cards/",
```

Dataset Update

- Website performance data collected through the Google PageSpeed Insights API.
- Data includes mobile and desktop results.

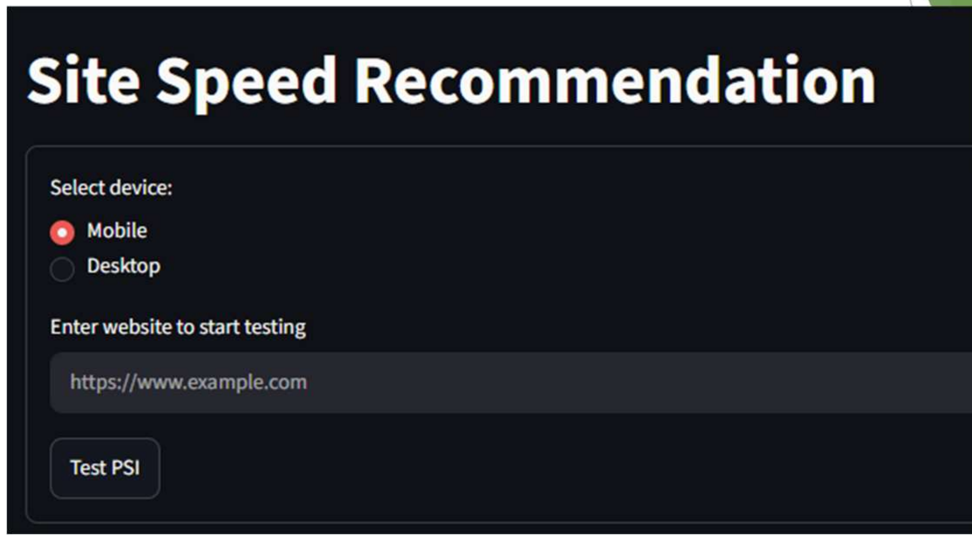
Dataset piece	Current status
Raw observations	~19,000 URL rows
Usable lab-data rows	~16,000 rows
Partial/missing lab rows	~3,000 rows
Decision needed	Use lab subset or rerun feature extraction for field data

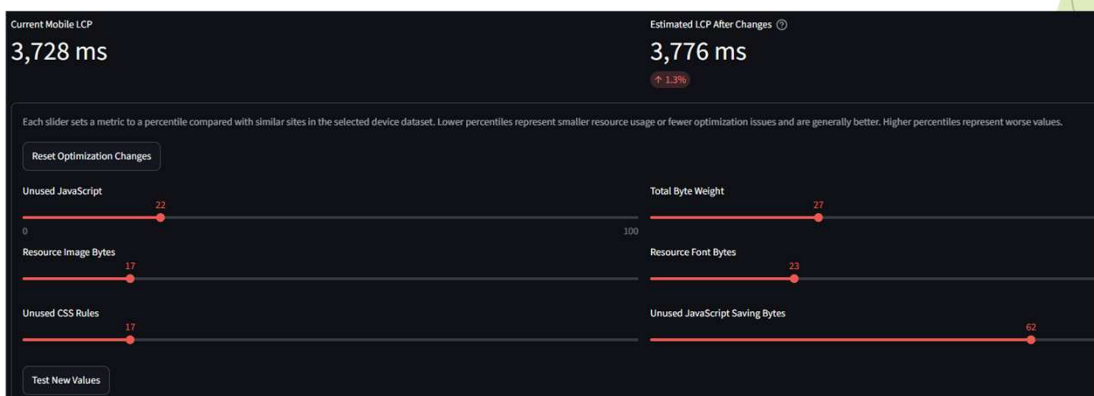
LCP Prediction Model

- Target variable: Largest Contentful Paint (LCP)
- Model: XGBoost Regression
- LCP was log-transformed during training to reduce the influence of extreme high-LCP values.
- Input features: PageSpeed Insight diagnostic metrics.

Dashboard Workflow

1. User enters a website URL and selects mobile or desktop testing.
2. The dashboard retrieves current PageSpeed Insights metrics.
3. Current LCP is displayed.
4. The user adjusts selected diagnostic metrics using percentile sliders.
5. The model estimates the new LCP and percentage change.





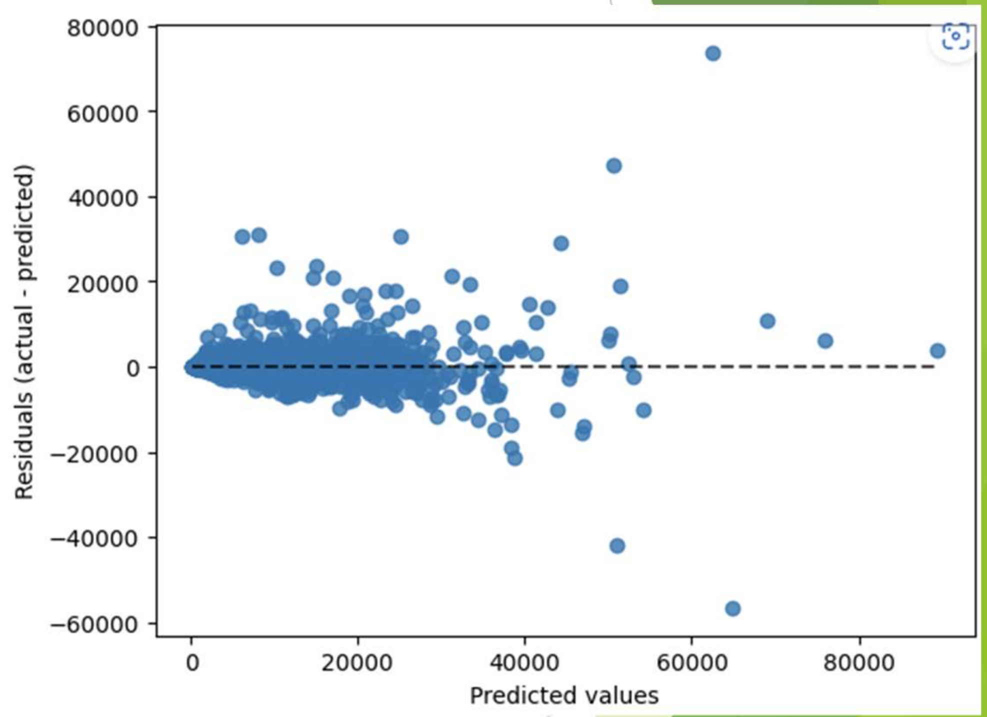
Model Evaluation

Metric	Role in project
Mean Absolute Error (MAE)	Primary model selection metric because the MAE isn’t skewed by outliers.
R-squared	Secondary metric used as a general measure of model performance
Recommendation validation	Generated recommendations will be compared against PageSpeed diagnostics.

Model Performance Results

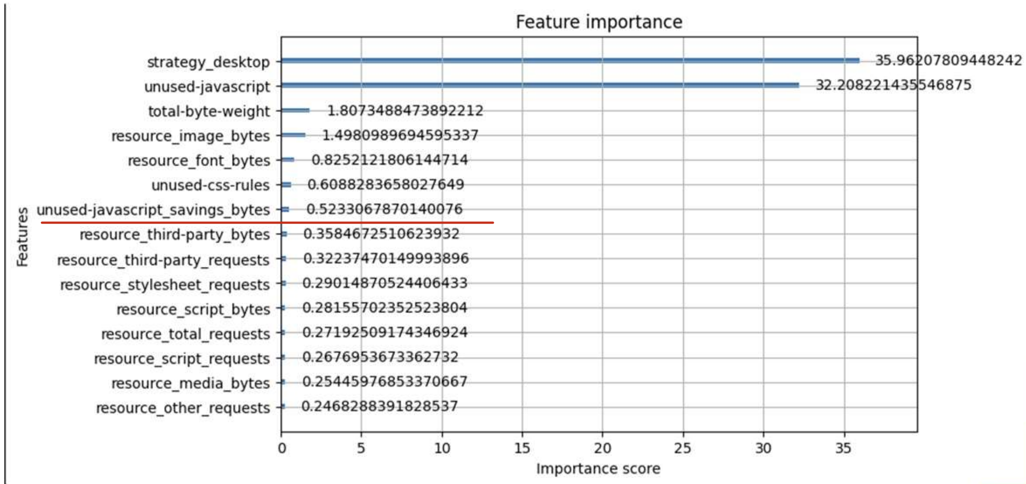
- Cross-validation and test results were in a similar range, indicating reasonable generalization.
- RMSE are higher than MAE because a small number of websites had extremely large LCP prediction errors.
- Residuals are centered near zero for most observations, but prediction error becomes more variable for extremely high LCP.

Metric	Result
Mean 5-fold CV MAE	1,541 ms
Test MAE	1,350 ms
Test RMSE	3,378 ms
Test R-squared	0.861



Feature Importance and Model Interpretation

- Device strategy and unused JavaScript were the most influential inputs in the trained model.
- Resource and diagnostic metrics provided small additional information.
- The dashboard allows users to adjust selected diagnostic inputs to simulate an estimated LCP.



Timeline

Phase	Status	Work Completed
1. Research question and website sample	Complete	Defined the original Core Web Vitals recommendation-system concept.
2. Data collection	Complete	Collected approximately 19,000 PageSpeed Insights observations.
3. Cleaning and EDA	Complete	Cleaned data, handled incomplete lab results, and retained approximately 16,000 usable rows.
4. Modeling	Complete	Tested regression approaches, addressed LCP outliers, selected XGBoost, and narrowed the scope to LCP.
5. Dashboard Development	Complete	Built the Streamlit dashboard with PSI testing, percentile-based adjustments, and estimated LCP output.
6. Final Evaluation and Delivery	Complete	Evaluated the model, documented limitations, and prepared the final presentation and report.

Limitations

- Current implementation predicts only LCP, not all Core Web Vitals.
- Website performance con vary by network conditions, device conditions, and changes to the live site.
- The current benchmark compares sites by device type. Category-based separations could be added later.

Future Work

1. Expand Performance Predictions
   1. Add models for the remaining CWVs: CLS and INP.
   2. Add supporting metrics like FCP, TTI, TBT.
   3. Provide a more complete view of website performance.
2. Add Category-Based Benchmarking
   1. Group websites into relevant categories, such as e-commerce, news, or education.
   2. Compare each website against the matching category and strategy.
3. Validate with Real Optimization Changes
   1. Add recommendation for users to fix website.

References

- Google PageSpeed Insights. https://pagespeed.web.dev/
- Google Web Vitals. https://web.dev/articles/vitals
- Google Lighthouse. https://developer.chrome.com/docs/lighthouse/